Daily Science Article

Topic: The First Digit Knows More Than It Should

by Nour Al-Zahraa Karnib October 4, 2026

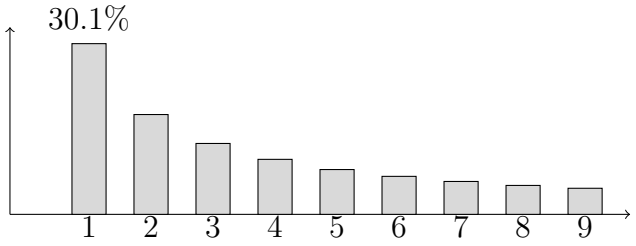
There is a strange habit hidden inside numbers.

It is so quiet that we usually never notice it. Take a collection of numbers: the populations of cities, the lengths of rivers, the areas of countries, the prices of houses, or measurements from a scientific experiment. Now ignore everything except the first digit.

At first, there seems to be no reason for one digit to appear more often than another. There are nine possible first digits, from 1 to 9, so we might expect each one to appear about equally often.

But nature does not seem to agree.

In many real-world collections of numbers, about 30% begin with the digit 1. The digit 2 appears first about 18% of the time, while 9 appears first only about 5% of the time.



At first, this sounds almost impossible. Why would numbers prefer to begin with 1?

This phenomenon is known as **Benford's Law**, and its explanation begins with a surprisingly simple change in perspective.

We normally think about numbers using addition. On an ordinary number line, the distance from 1 to 2 is exactly the same as the distance from 8 to 9.

But many things in the real world do not grow by adding the same amount. They grow by multiplying.

Going from 1 to 2 means multiplying by 2. Going from 10 to 20 is also multiplying by 2. Going from 100 to 200 is again multiplying by 2, even though the numerical differences are completely different.

The first digit is therefore connected more naturally to *scale* and *ratios* than to ordinary distance.

This is where logarithms quietly enter the story.

Consider all numbers whose first digit is 1. They begin at 1 and continue until just before 2. On a logarithmic scale, that interval has length

$$\log_{10}(2) - \log_{10}(1) = \log_{10}(2).$$

Since

$$\log_{10}(2) \approx 0.301,$$

the first digit 1 occupies about 30.1% of one complete logarithmic scale.

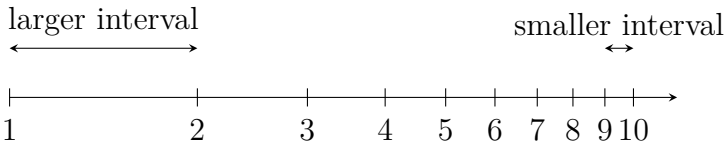
That is exactly where the strange 30% comes from.

The same reasoning works for every digit. If the first digit is d, the probability predicted by Benford's Law is

$$P(d) = \log_{10}\left(1 + \frac{1}{d}\right).$$

For $d = 1$, the probability is about 30.1%. For $d = 9$, it is only about 4.6%.

So the digits were never really competing on equal terms.



There is another beautiful feature hiding inside the law.

Imagine multiplying every number in a collection by 10. The first digits do not change. The number 13 becomes 130, 27 becomes 270, and 91 becomes 910. Their first significant digits remain exactly the same.

This property is called **scale invariance**.

It explains why a similar pattern can appear in completely different kinds of data. Populations, geographical measurements, financial quantities, and physical measurements can all span several orders of magnitude, and when

they do, the logarithmic structure of their values can reveal this unexpected first-digit pattern.

But Benford's Law is not a rule that every list of numbers must obey. If we simply write the numbers from 1 to 1000, the pattern does not automatically appear. The data need to have the right kind of distribution and usually need to spread across several different scales.

And that is what makes the idea more interesting.

The mathematics is not really saying that the digit 1 is special.

It is saying that **our ordinary way of looking at numbers can sometimes hide their real structure**.

On a normal number line, 1 to 2 and 8 to 9 look equally large. On a logarithmic scale, however, they are not equal. The interval from 1 to 2 occupies much more space than the interval from 9 to 10.

The digit 1 therefore appears first more often not because numbers "like" 1, but because multiplication and scale have changed the geometry of the problem.

A strange numerical coincidence becomes a predictable mathematical pattern simply by changing the way we look at numbers.

And perhaps that is the most beautiful lesson hidden inside Benford's Law:

Sometimes mathematics does not change the world we are looking at.

It simply changes the way we look at it.